1 Setup for the lecture

The following document will contain some setup recommendations for the lecture.

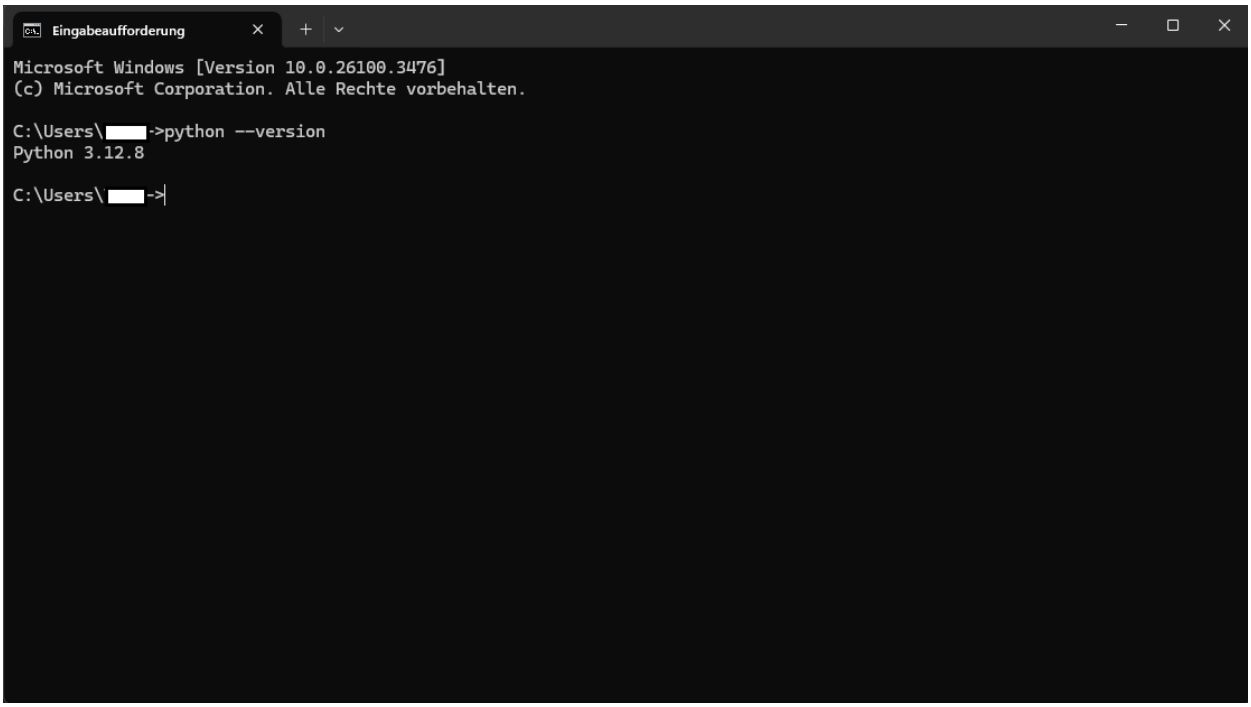
2 Installing Python

If you are not familiar with Python the following tutorial is recommended:
https://jensliebehenschel.github.io/ShortPythonIntro/en/

Download Python through their official website:
https://www.python.org/downloads/

Open the File.
Tick 'Add python.exe to PATH'.
This will add python to your environment variables.

Verify the installation by opening the command prompt and write ′python –version′



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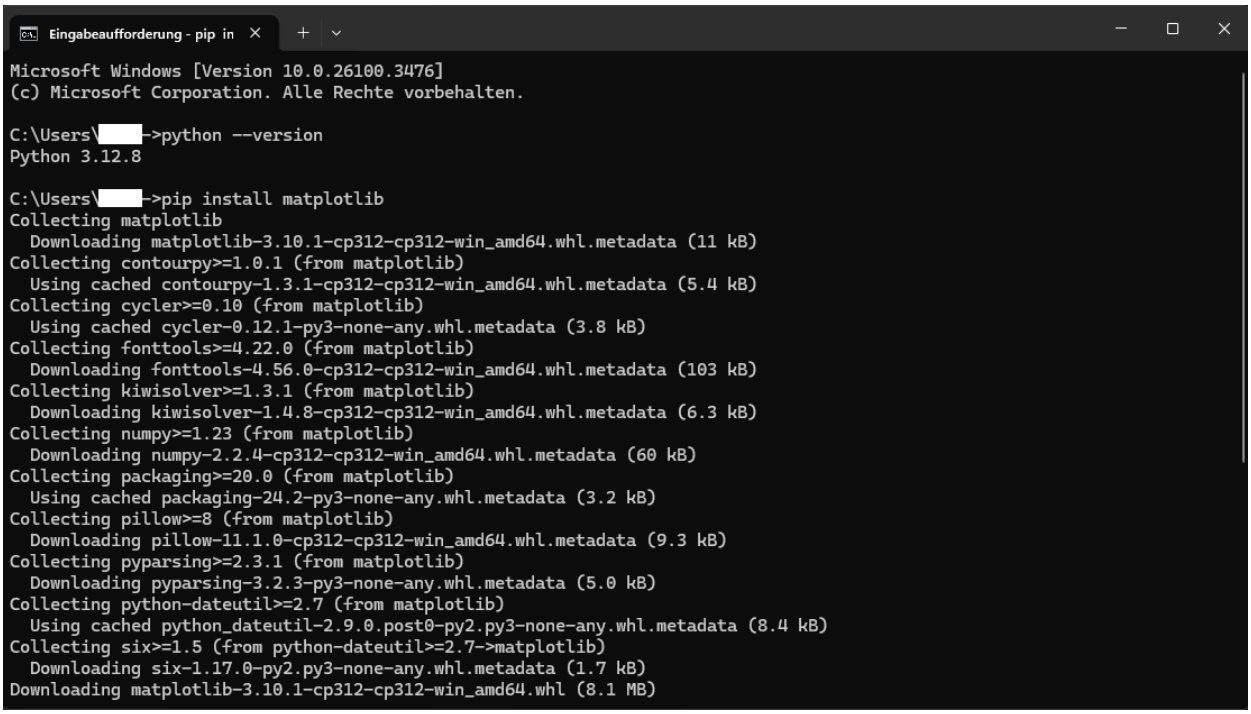
3 Installing Packages

Variant 1: pip (under Windows pip3)
You can install packages for your whole system via 'pip install <package-name>' in your console.



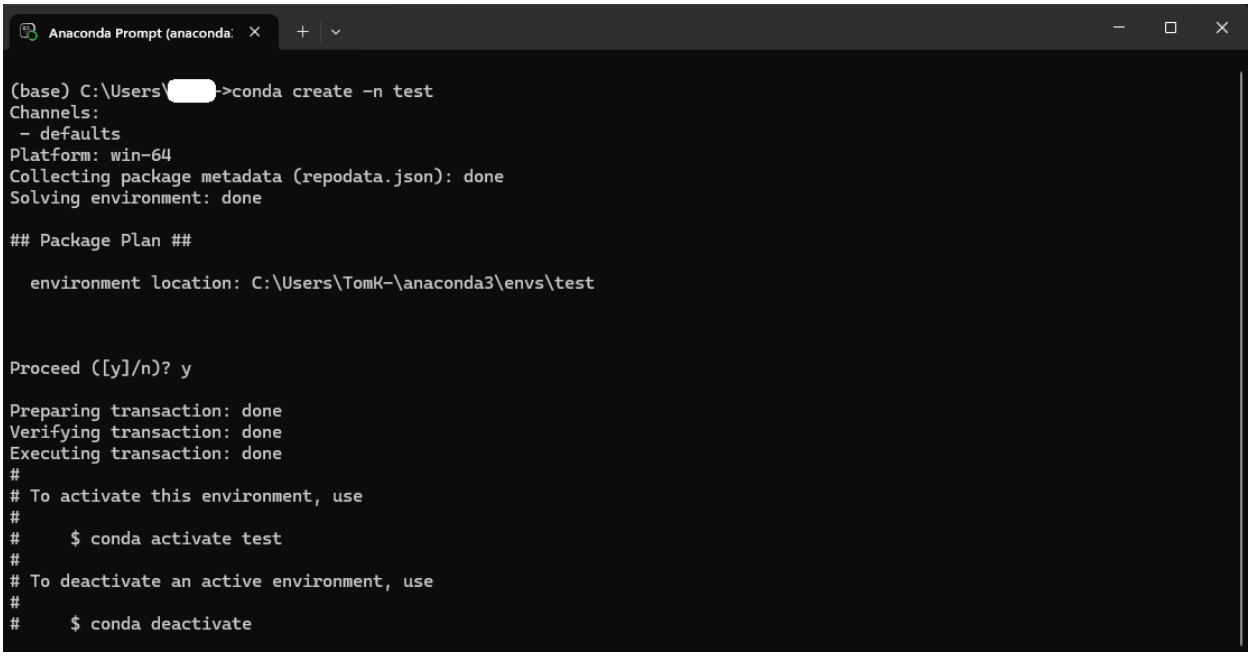
```
Eingabeaufforderung - pip in
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. Alle Rechte vorbehalten.

C:\Users\[redacted]->python --version
Python 3.12.8

C:\Users\[redacted]->pip install matplotlib
Collecting matplotlib
  Downloading matplotlib-3.10.1-cp312-cp312-win_amd64.whl.metadata (11 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Using cached contourpy-1.3.1-cp312-cp312-win_amd64.whl.metadata (5.4 kB)
Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib)
  Downloading fonttools-4.56.0-cp312-cp312-win_amd64.whl.metadata (103 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.4.8-cp312-cp312-win_amd64.whl.metadata (6.3 kB)
Collecting numpy>=1.23 (from matplotlib)
  Downloading numpy-2.2.4-cp312-cp312-win_amd64.whl.metadata (60 kB)
Collecting packaging>=20.0 (from matplotlib)
  Using cached packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
Collecting pillow>=8 (from matplotlib)
  Downloading pillow-11.1.0-cp312-cp312-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=2.3.1 (from matplotlib)
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Collecting python-dateutil>=2.7 (from matplotlib)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.7->matplotlib)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading matplotlib-3.10.1-cp312-cp312-win_amd64.whl (8.1 MB)
```

Variant 2: conda
With conda you create an environment e.g. 'AI-Module'.
The packages you install using conda are only installed for the specific environment and not for the whole system.
This is helpful if you work on multiple projects at the same time since you can have multiple version of the same package installed, and it is easy to share environments with others.
Step 1: Go to https://www.anaconda.com/download/success and download Anaconda (or miniconda if you prefer a lightweight variant) for your operating system.
Step 2: Open the file and install the file (Tip: Add conda to your environment variables).

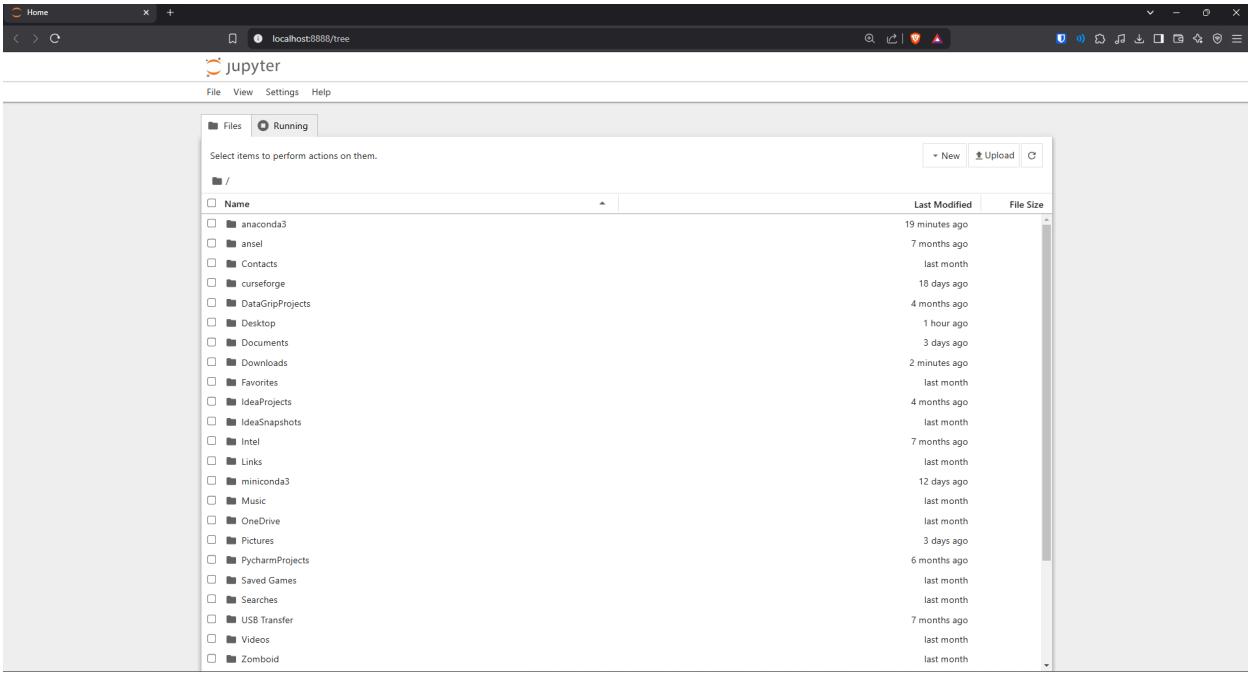
Installing packages with conda:
Step 1: Open the Anaconda Prompt application
Step 2: Create an environment with the command 'conda create -n <environment-name>'



Step 3: Activate the environment with the command 'conda activate <environment-name>'
Step 4: packages can be installed via 'conda install <package-name>'

4 Installing Jupyter Notebook

If you use Anaconda just open the Anaconda Navigator. There you will find a launch option for Jupyter Notebooks.
If you do not use Anaconda install Jupyter Notebooks via 'pip install jupyter'. After the installation run jupyter notebook and an application should open in your browser.



Tip: PyCharm has quality of life features when it comes to using jupyter notebooks. However, you will probably be prohibited from using PyCharm in the Exam

5 Very Rudimentary Plot Tutorial

For the creation of plots, we commonly use the matplotlib.pyplot python package.
For a more detailed description please reference the official package documentation : https://matplotlib.org/stable/api/pyplot_summary.html

Common functions:

- import matplotlib.pyplot as plt
- plt.plot(x, y) is used to create a simple scatterplot with the arguments passed.
- plt.set_xlabel(string) is used to add a name to the x-axis.
- plt.set_ylabel(string) is used to add a name to the y-axis.
- plt.set_title(string) is used to add a title to the plot.
- plt.xlim(left, right) is used to limit the x-axis.
- plt.ylim(top, bottom) is used to limit the y-axis.
- plt.tight_layout() is used for lazy formatting.
- plt.legend() is used to display the labels set in plt.plot()
- plt.show() is used to display the plot in its current state.

Example (Simple Scatterplot):

```
import matplotlib.pyplot as plt

x = <x-axis data>
y = <y-axis data>
plt.plot(x, y)
plt.show()
```

To create more complex plots other packages can be used, such as the python package seaborn.

Example (Heatmap):

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.heatmap(<data>.corr(), annot=True, cmap='coolwarm', fmt='.2f')\\
plt.show()
```

Example (Pairplot):

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.pairplot(<data>, kind='kde')
plt.show()
```